

**The Second Law of
Thermodynamics is bent but not
broken**

Module-4: Second Law of Thermodynamics

Thermodynamics (NMEC103) : (Winter 2024-25)

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Limitations of 1st Law of Thermodynamics.

- First law of Thermodynamics is a law of conservation of total energy (Mechanical, Thermal, Chemical Nuclear etc.), which gives a quantitative estimate of heat and work interaction between system and surrounding.
- It does not specify the direction of a process. All spontaneous process proceeds in one direction only.
- The first law of thermodynamics does not deny the feasibility of a process reversing itself.

1st Law of Thermodynamics does not provide answer to the following questions

- ✓ Is a particular process/ reaction feasible?
- ✓ To what extent does the process/ reaction proceed?
- ✓ Is complete conversion of internal energy into work possible?
- ✓ What is the maximum limit of energy conversion?

Questions Unanswered by 1st law of Thermodynamics

- If **energy is conserved**, why we are worried for **conservation of energy**?
- Is there any **preferred direction** for the **spontaneous process**?
- Is there any **maximum limit** (other than infinity) for the performance indicator of heat engine or a device?
- Can **heat be completely converted into work**?
- Establishing the **condition of equilibrium** (Rusting of Iron).
- **Quantitative evaluation** of the factors that prevent the attainment of the best performance.
- Is there **states** which can not be achieved by a system **adiabatically**?

2nd Law of Thermodynamics is the

- Law of *direction*
- Law of *quality*
- Law of *limits*
- Law of *equality*
- Law of *inequality*

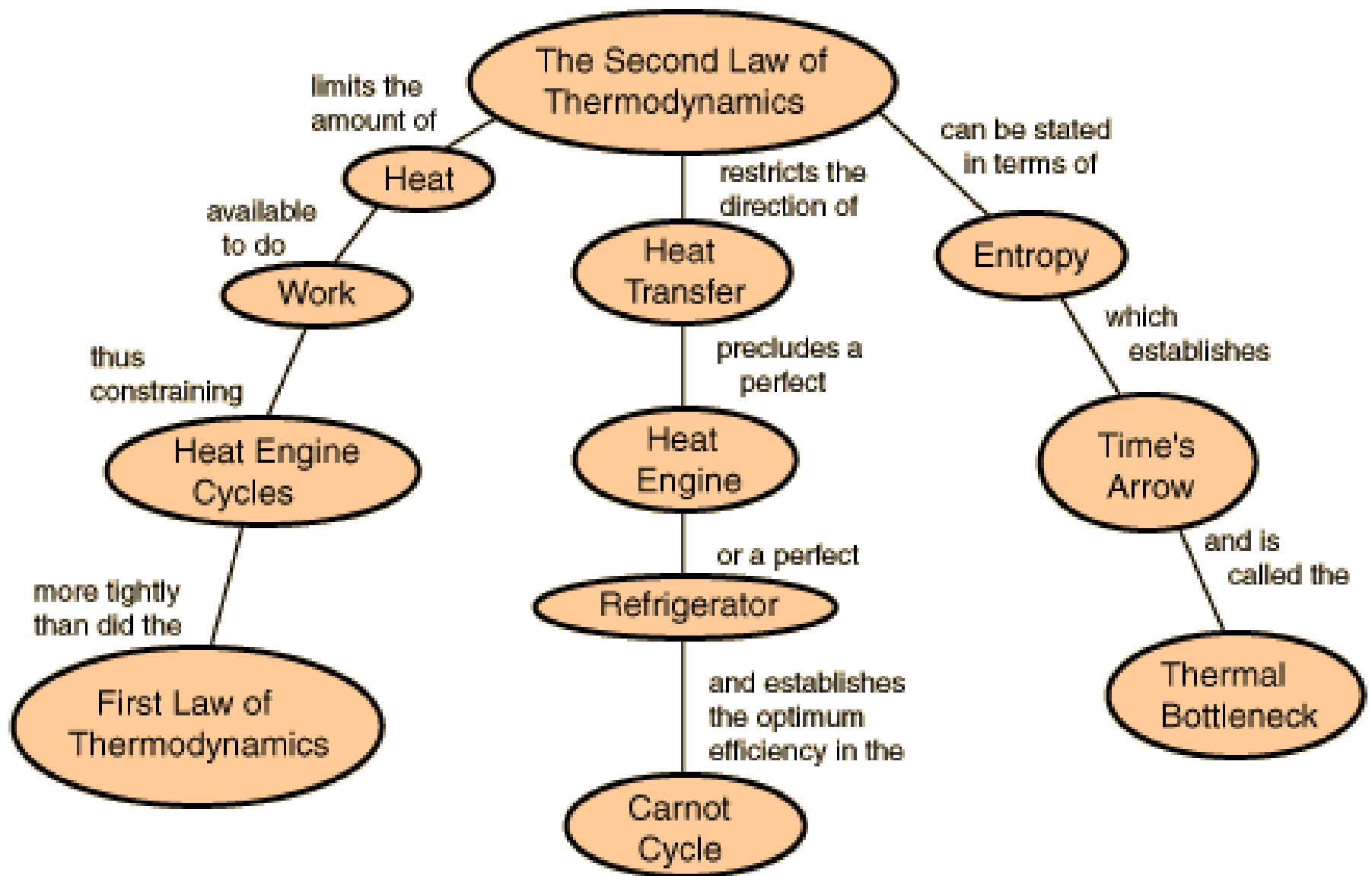
2nd Law of Thermodynamics

- Provides a means for *establishing condition for equilibrium*
- Useful in defining an *absolute temperature scale* independent of thermometric substance
- Useful in *evaluating the properties* that can not be measured directly (e.g. u , h , s)

❖ First law denies the possibility of creating or destroying energy.

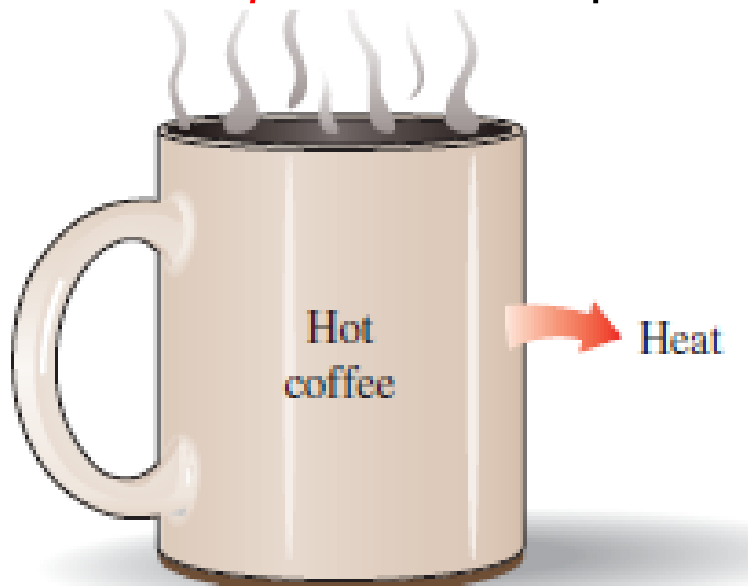
❖ The Second Law denies the possibility of using energy in a particular direction

- 1st Law gave Birth to a Property Called Energy (Total energy of the Universe is Constant)
- 2nd Law gave Birth to a Property Called Entropy (Total Entropy of the Universe is Increasing)

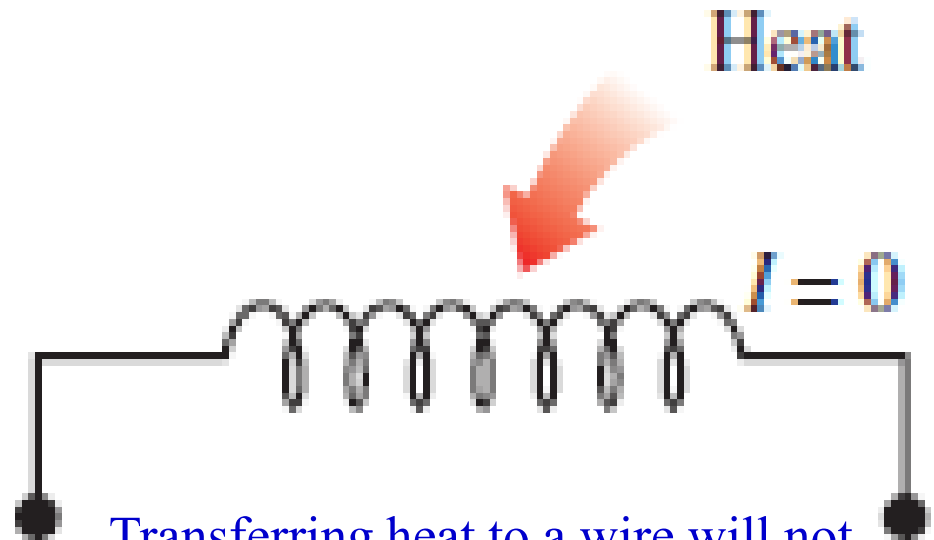


Second Law of Thermodynamics

Law of Direction: Second law ascertains the **direction of any process** and **discards the impossible process** by introducing a new property called **entropy**. A process will occur in a direction such that the **total entropy of the universe** (i.e. System+ Surrounding) **must increase**. In other words the **total entropy generation due to the process must be either positive or zero in the limiting case (Complete Reversible Process)**. It is a matter of everyday experience that there is a definite direction for **spontaneous** processes (irreversible in nature).



A cup of hot coffee does not get hotter in a cooler room.

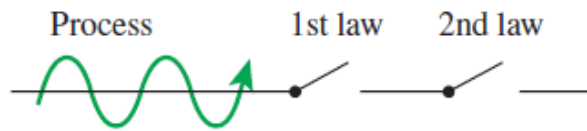
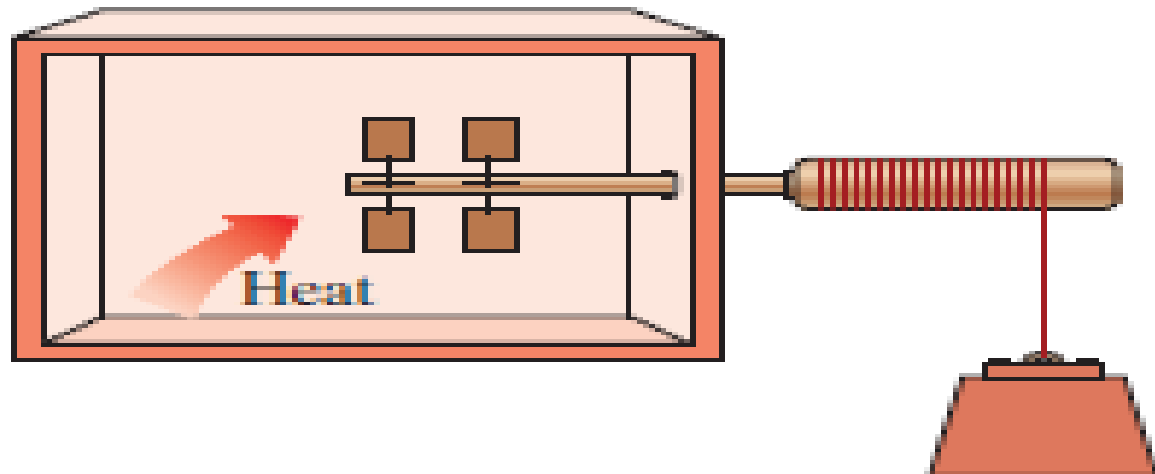


Transferring heat to a wire will not generate electricity.

Impossible Processes

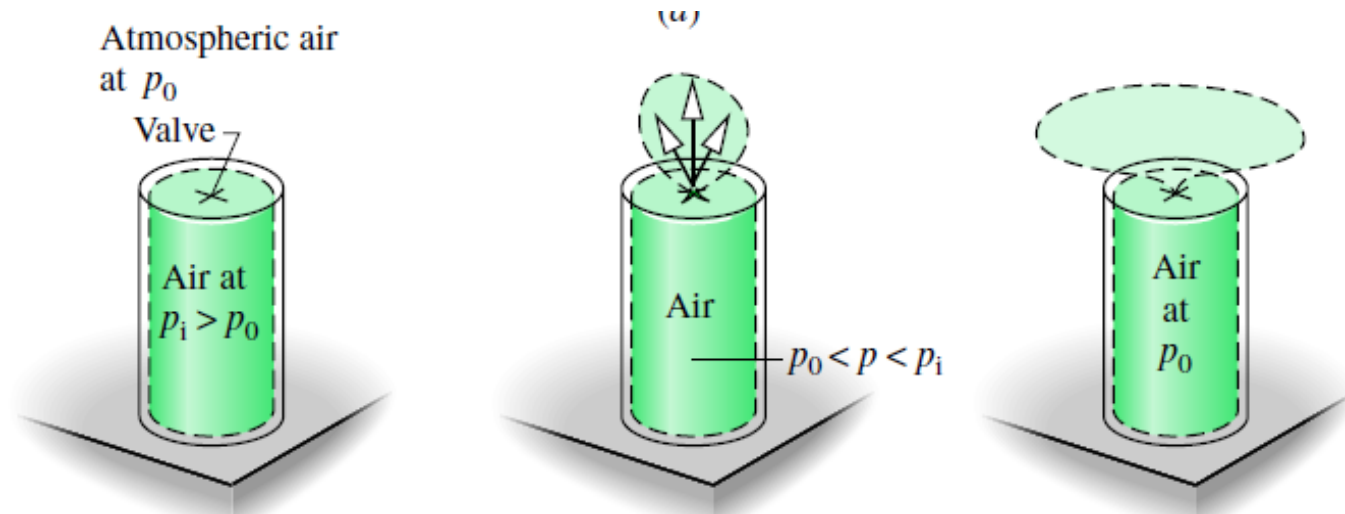


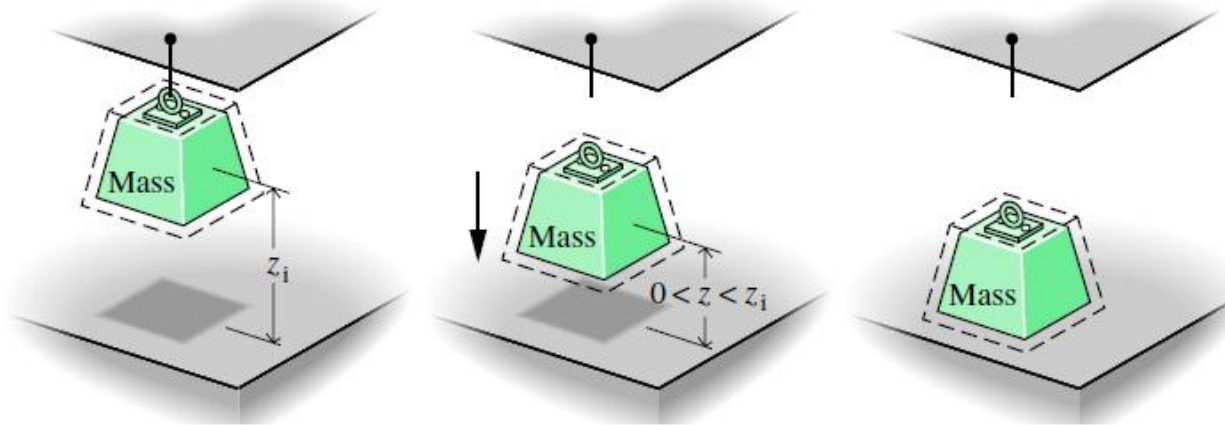
Process occurs in a certain direction, and not in the reverse direction



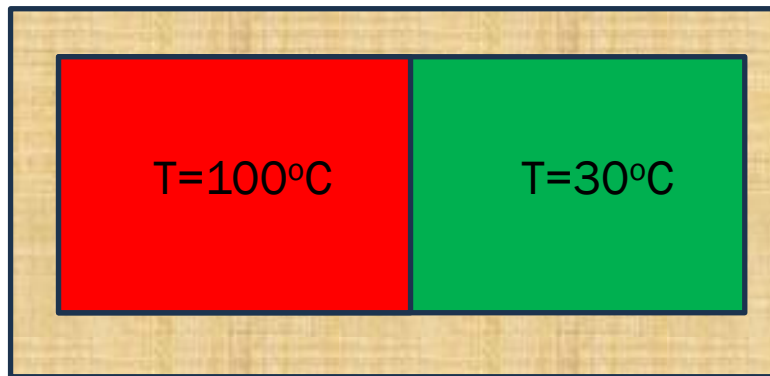
Transferring heat to a paddle wheel will not cause it to rotate.

A Process must satisfy 1st and 2nd law of thermodynamics to proceed



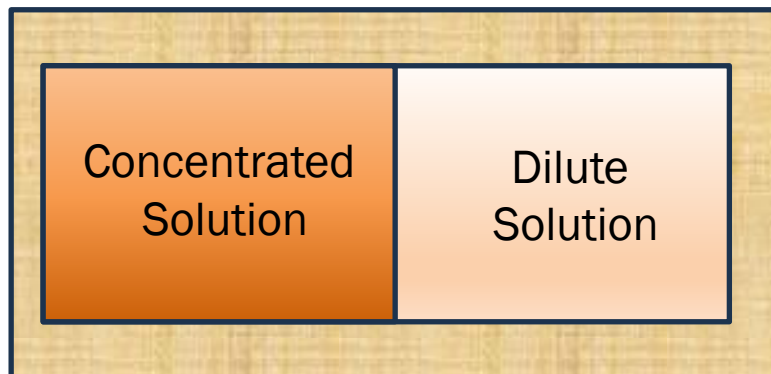


spontaneous processes



$$\Delta U = 0 \quad 30^{\circ}C < T_f < 100^{\circ}C$$

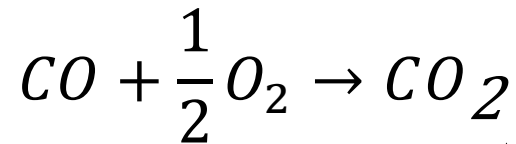
The **magnitude of ΔU** is also zero if **nothing is happening**. Hence the magnitude and direction of ΔU has no indication towards the direction of spontaneous process



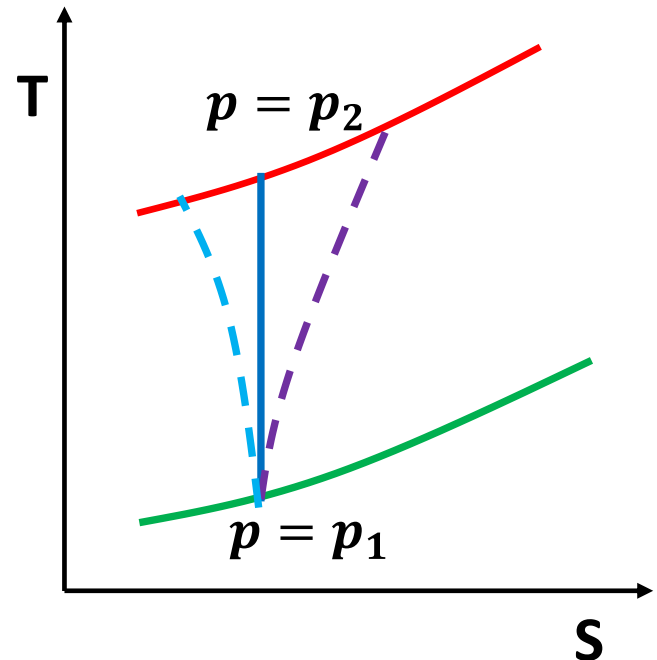
$$\Delta U = 0 \quad C_{dil} < C_f < C_{con}$$

➤ Steel expands, when heated, when it is stretched adiabatically within the elastic limits does its temperature increase or decrease ?

➤ Sixteen kilograms of oxygen and 28 kg of carbon monoxide fill a tank. The mixture is heated until it reaches 250 kPa, 500 °C to what extent does the following reaction occur?



➤ Is it possible to compress air adiabatically from 100 kPa, and 15 °C ($s = 5.665$ kJ/kg-K) to 200 kPa, 30 °C ($s = 5.517$ kJ/kg-K)?



- Ideal gas is flowing through an adiabatic opaque pipe (you can not see the flow). The inlet and exit pressure and temperature given. Can you guess the direction of flow?

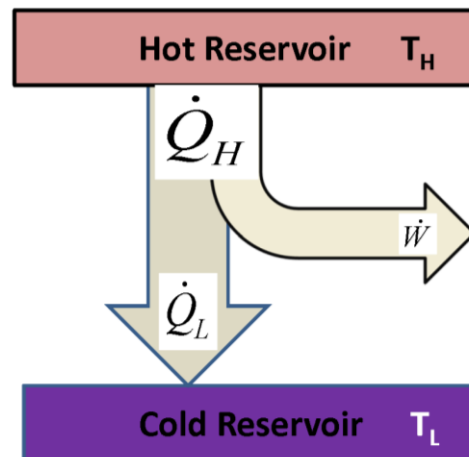


As the entropy generation associated with a reversible process is zero, a reversible process can proceed in either direction without violating 2nd law of thermodynamics. However, an irreversible process can proceed in a particular direction in which the **entropy generation is +ve** (i.e., the 2nd law) not in the reverse direction (**entropy generation is -ve not possible**).

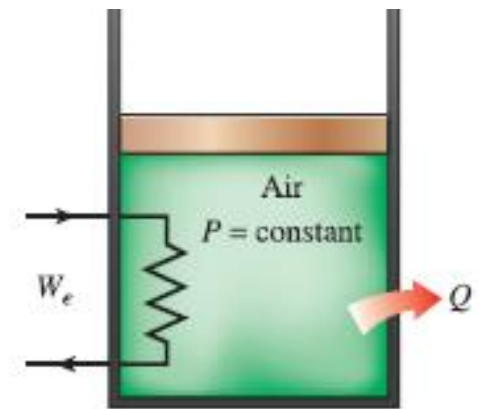
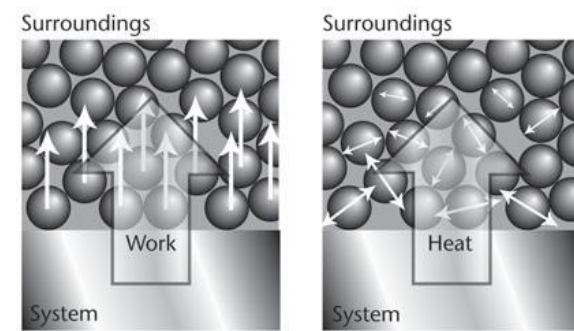
Law of Quality:

According to 2nd Law energy has both **quantity** as well as **quality** (Work Potential). Although **energy is conserved quantitatively** (1st Law), it **degrades qualitatively**, hence when we are bombarding with the statement that “Conserve Energy” we **mean the conservation of quality of the energy not the quantity** (as it is always conserved)

If the heat engine shown in the figure is a **fully reversible both externally and internally** (say **Carnot Heat engine**, but **it is not mandatory that it should be Carnot**) and **the heat sink is at the ambient temperature** (i.e. $T_L = T_0$) then the work obtained is the maximum possible work that can be extracted from Q_H at T_H in an **environment at T_L** , also known as **quality** or the work potential of heat Q_H at T_H .



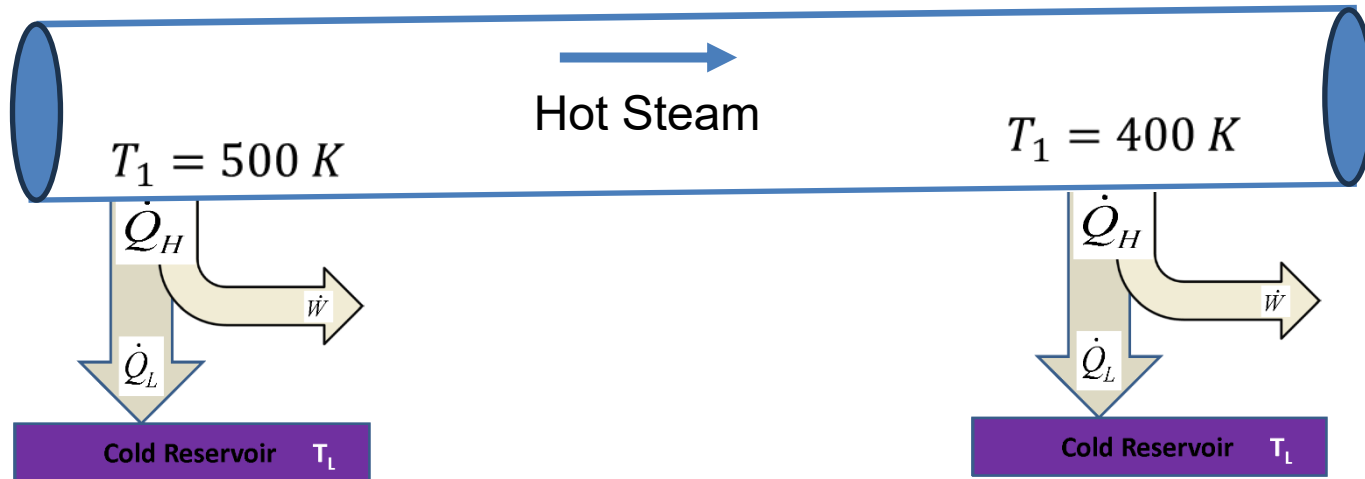
According to 2nd law heat is a low grade energy (because of the) disordered motion of the molecules and work (needs ordered motion e.g. electric work, electrons are moving in a particular direction, moving boundary work, molecules follow the direction of movement of the piston) is a high grade energy. Second law put a constraint on the complete conversion of heat (low grade energy) to work (high grade energy). According to 2nd law heat can be completely converted in to work (in a process it is possible. e.g. Isothermal expansion of ideal gas) but not continuously (i.e. in a cycle it is not possible). If the above heat engine were a completely reversible one, the heat lost to the cold reservoir i.e. Q_L is not because of any dissipative effect or inefficiency of the heat engine rather due to constraint put by the second law of thermodynamics



2nd law does not put any constraint on conversion of high grade to low grade energy

1st law – internal energy (quantity of energy)

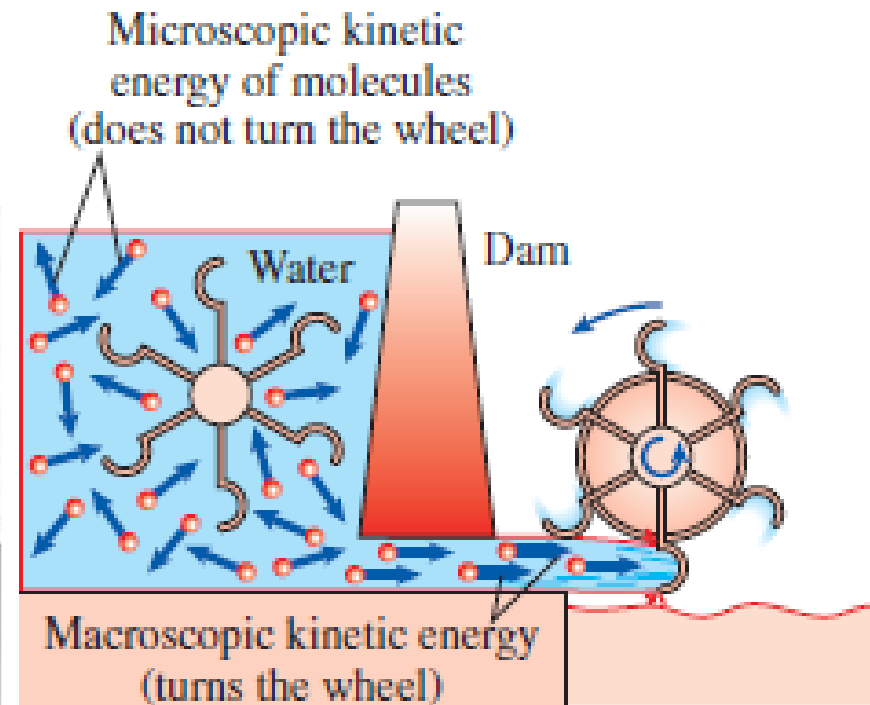
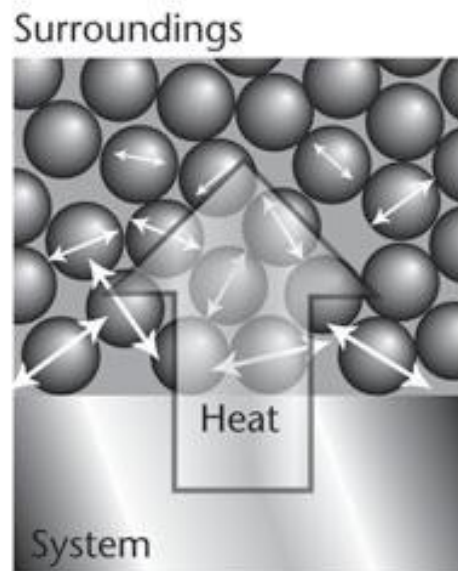
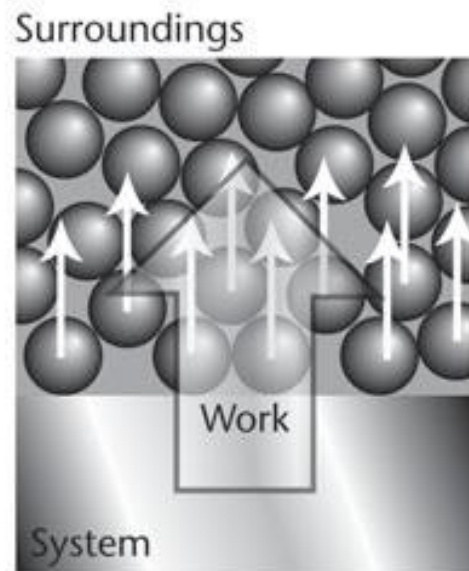
2nd law – Entropy (quality of energy)



$$W_C = Q_H \left(1 - \frac{300}{500} \right) = 0.4 Q_H$$

$$T_L = 300\text{K} \quad W_C = Q_H \left(1 - \frac{300}{400} \right) = 0.25 Q_H$$

According to 2nd law heat is a low grade energy (because of the) disordered motion of the molecules and work (needs ordered motion e.g. electric work, electrons are moving in a particular direction, moving boundary work, molecules follow the direction of movement of the piston) is a high grade energy.



High Grade Energy

- ☐ Mechanical Work
- ☐ Electrical Energy
- ☐ Water Power
- ☐ Wind power
- ☐ Kinetic energy of a jet
- ☐ Tidal Power

Low Grade Energy

- ☐ Heat and Thermal Energy
- ☐ Heat derived from Nuclear fission and fusion
- ☐ Heat derived from combustion of fossil fuel

Dissipative Effect

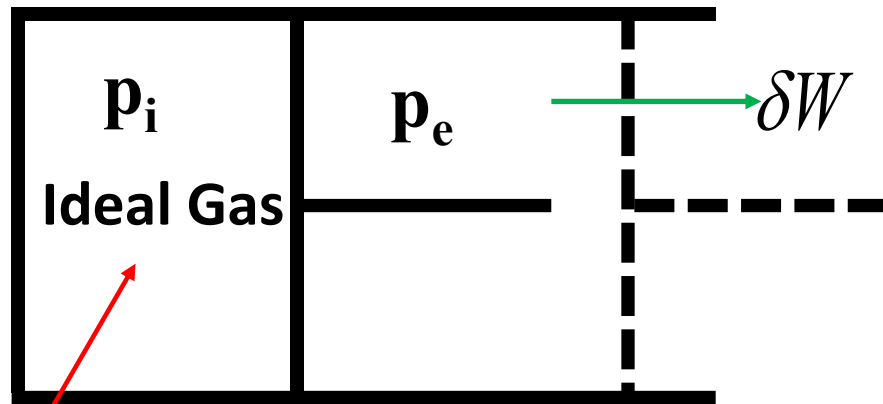
The diagram illustrates the flow of energy between two states. A green arrow points from the 'High Grade Energy' box to the 'Low Grade Energy' box, labeled 'Dissipative Effect'. A red arrow points from the 'Low Grade Energy' box back to the 'High Grade Energy' box, labeled 'Constrained by 2nd Law'. The red arrow is positioned below the green arrow, indicating that the reverse process is constrained by the second law of thermodynamics.

Constrained by 2nd Law

Second law put a constraint on the complete conversion of **heat** (low grade energy) **to work** (high grade energy).

Can Heat be Completely Converted into Work?

e.g. : Isothermal Expansion of an Ideal Gas



However, this process stops when $p_i = p_e$ (T is constant as v increases p decreases)

$\delta Q = dU + \delta W$ 1st Law for **closed stationary system**

$dU = C_v dT$ For **ideal gas** undergoing any process

$\delta Q = \delta W$ For **ideal gas** undergoing **isothermal expansion**

Heat can be **completely** converted into work in a **process**.

What about cycle?

Statements of Second Law of Thermodynamics

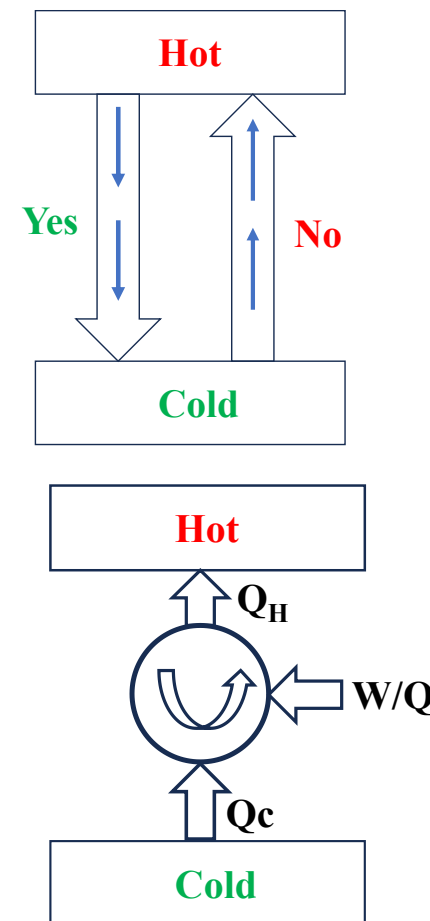
Clausius' statement of second law (1850):

It is impossible for any system to operate in such a way the sole result would be an energy transfer by heat from a cooler to hotter body

The Clausius statement does not rule out the possibility of transferring energy by heat from a cooler body to a hotter body, for this is exactly what refrigerators and heat pumps accomplish. However, as the words “sole result” in the statement suggest, when a heat transfer from a cooler body to a hotter body occurs, there must be *other effects* within the system accomplishing the heat transfer, its surroundings, or both



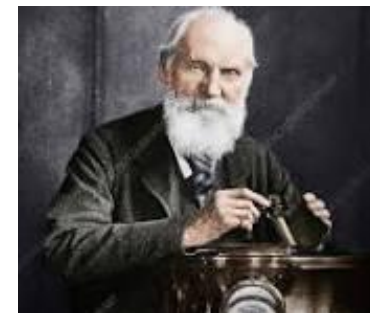
Clausius



Statements of Second Law of Thermodynamics

Kelvin-Planck statement of second law (1897):

It is impossible to construct a device (engine) **operating in a cycle** that will produce **no effect other than extraction of heat from a single reservoir** and **convert all of it into work**. The Kelvin-Planck statement of the 2nd law precludes a heat engine producing work which is receiving heat from a single thermal reservoir i.e. the net work for the cycle can not be positive. **However, there is no restriction on net work input (net work is negative) or zero work for the closed system cycle.** Thus the analytical expression for Kelvin-Planck's Statement of 2nd Law:



Kelvin

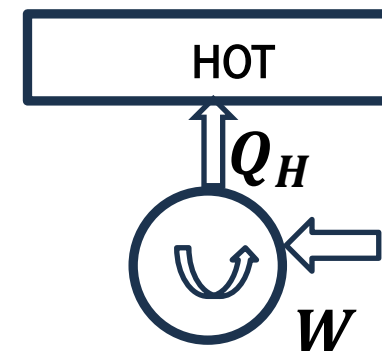
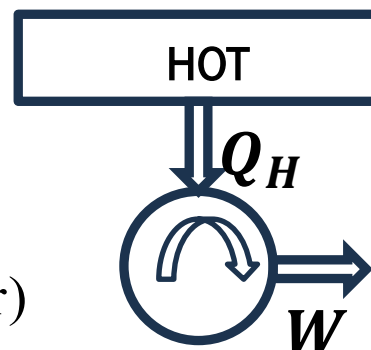


Max Planck

$$\oint \delta W_{closed} \leq 0 \text{ (For a single thermal reservoir)}$$

For thermodynamic cycle $\oint \delta Q = \oint \delta W$

Hence $\oint \delta Q \leq 0$ (For a single thermal reservoir)



A system **undergoing a thermodynamic cycle** interacting with a single thermal reservoir will undergoes **net negative heat and work interaction** (Statement of 2nd law)

Statements of Second Law of Thermodynamics

Irreversible Cycle

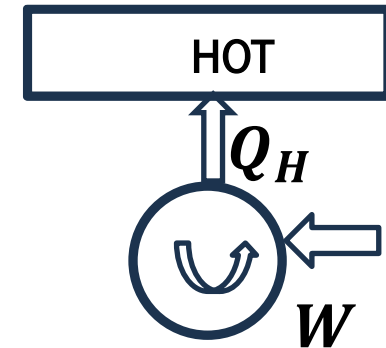
$$\oint \delta W_{closed} < 0 \text{ (For a single thermal reservoir)}$$

$$\text{Hence } \oint \delta Q < 0 \text{ (For a single thermal reservoir)}$$

Reversible Cycle

$$\oint \delta W_{closed} = 0 \text{ (For a single thermal reservoir)}$$

$$\text{Hence } \oint \delta Q = 0 \text{ (For a single thermal reservoir)}$$



No reversible cycle is possible with single reservoir

Imp.Note: If this reversible cycle were possible the reverse of this must have been possible. Which will violate the 2nd law. Hence No reversible cycle is possible with single thermal reservoir.

Analytically : For cyclic device delivering a net work output by extracting heat with a single reservoir

$$\text{From 1}^{\text{st}} \text{ Law : } \oint \delta Q = \oint \delta W$$

$$\Rightarrow Q_H = W$$

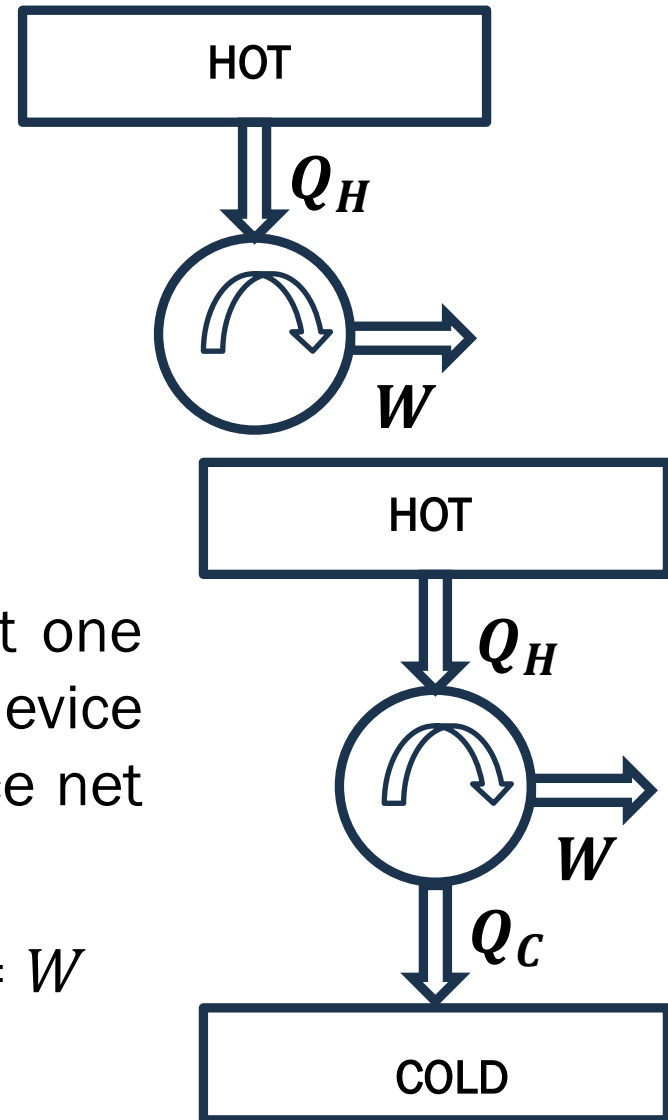
$$\therefore \text{Efficiency, } \eta_{th} = \frac{W}{Q_H} = 100\%$$

However, this ruled out by 2nd law

According to 2nd law there must be at least one more cold reservoir to which the cyclic device must reject some amount of heat to produce net work output

$$\text{From 1}^{\text{st}} \text{ Law : } \oint \delta Q = Q_H - Q_L = \oint \delta W = W$$

$$\therefore \text{Efficiency, } \eta_{th} = \frac{W}{Q_H} = \frac{Q_H - Q_L}{Q_H} = 1 - \frac{Q_L}{Q_H} < 100\%$$



Clausius' statement of second law (1850):

It is impossible to construct a heat pump or refrigerator which can pump heat from a low temperature reservoir to a high temperature reservoir without any work input, while operating in a cycle. Alternatively, "It is impossible to construct a heat pump or refrigerator which has infinite COP".

Kelvin-Planck statement of second law (1897):

It is impossible construct a heat engine with 100% thermal efficiency

Entropy Statement of the Second Law

Mass and energy are familiar examples of extensive properties of systems. Entropy is another important extensive property. Just as mass and energy are accounted for by mass and energy balances, respectively, entropy is accounted for by an entropy balance. In words, the entropy balance states

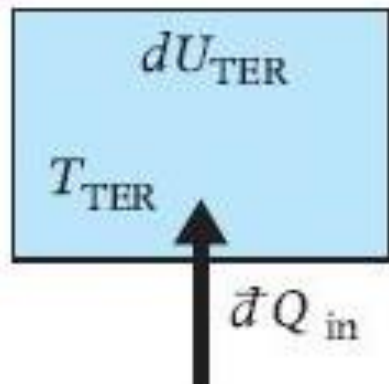
$$\left[\begin{array}{c} \text{Change in the amount of entropy} \\ \text{contained within the system in} \\ \text{a given time interval} \end{array} \right] = \left[\begin{array}{c} \text{Net amount of entropy} \\ \text{transferred} \\ \text{in across the system boundary} \\ \text{during} \\ \text{the time interval} \end{array} \right] + \left[\begin{array}{c} \text{Amount of entropy} \\ \text{produced} \\ \text{during the time interval} \end{array} \right]$$

Unlike mass and energy, which are conserved, entropy is produced (or generated) within systems whenever nonidealities (called irreversibilities) such as friction are present. The entropy statement of the second law states: *It is impossible for any system to operate in a way that entropy is destroyed.*

Thermal Energy Reservoir (TER)

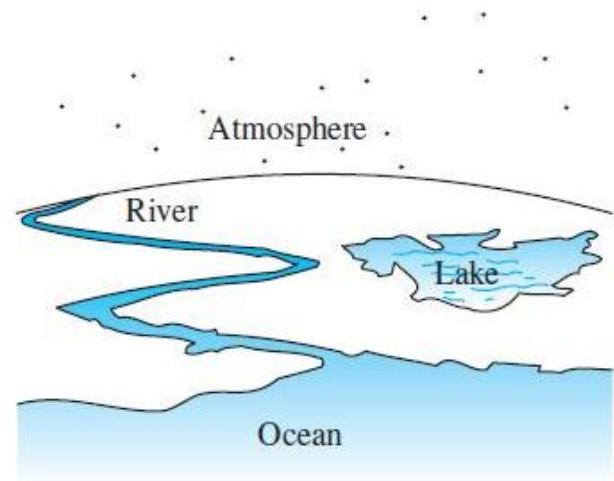
Thermal energy reservoir is imagined as a large mass (*high thermal capacity* mc) of some simple compressible substance and is in **internal equilibrium at a uniform and constant temperature T** , even though energy is added or extracted from the system. **The temperature is the identity of a TER.** TER act as an energy source or sink for energy transfer as heat. TER always undergoes at least internally reversible process. The Internal energy of a TER may change but not the temperature .

Ex. **Earth's atmosphere**, **Large water bodies** (sea, rivers etc.) **Large copper blocks** etc. A system undergoing phase change is also considered as thermal energy reservoir

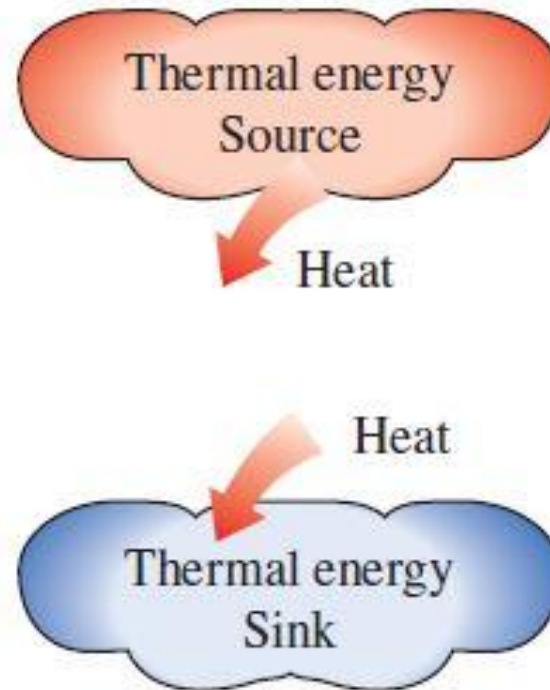


$$dS = \delta Q_{\text{in}} / T_{\text{TER}}$$

$$dU = mcdT$$

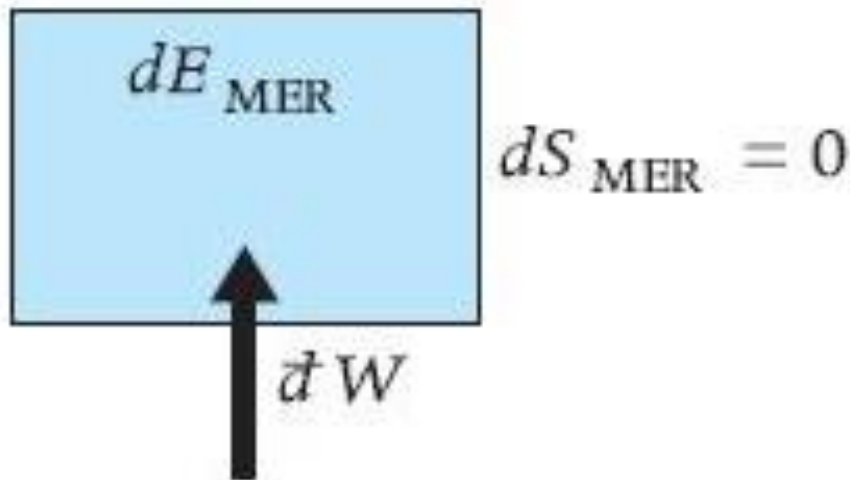


A reservoir that supplies energy in the form of heat is called a **source**, and one that absorbs energy in the form of heat is called a **sink** (Fig. 6–7). Thermal energy reservoirs are often referred to as **heat reservoirs** since they supply or absorb energy in the form of heat.



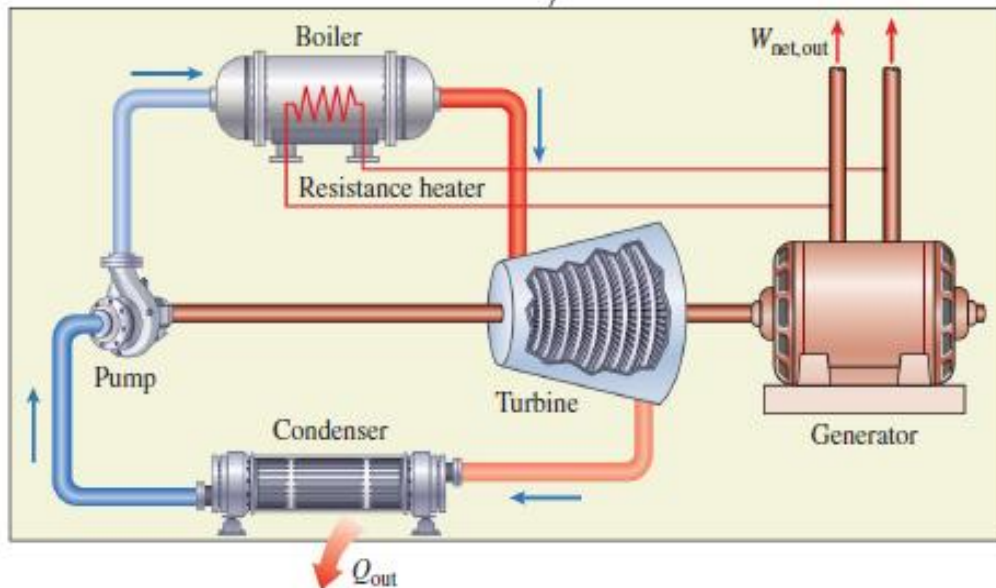
Mechanical Energy Reservoir (MER)

A mechanical energy reservoir (MER) is imagined as a frictionless flywheel that can store energy, it receives as work in a microscopically fully organized manner (molecules rotating around the axis in tandem), and return that energy without any loss through the output of energy as work (see Figure). There is no change in the entropy (microscopic disorder) in a MER when it receives or delivers energy, so by concept. MER always undergoes internally reversible process



Perpetual-Motion Machines of 1st kind

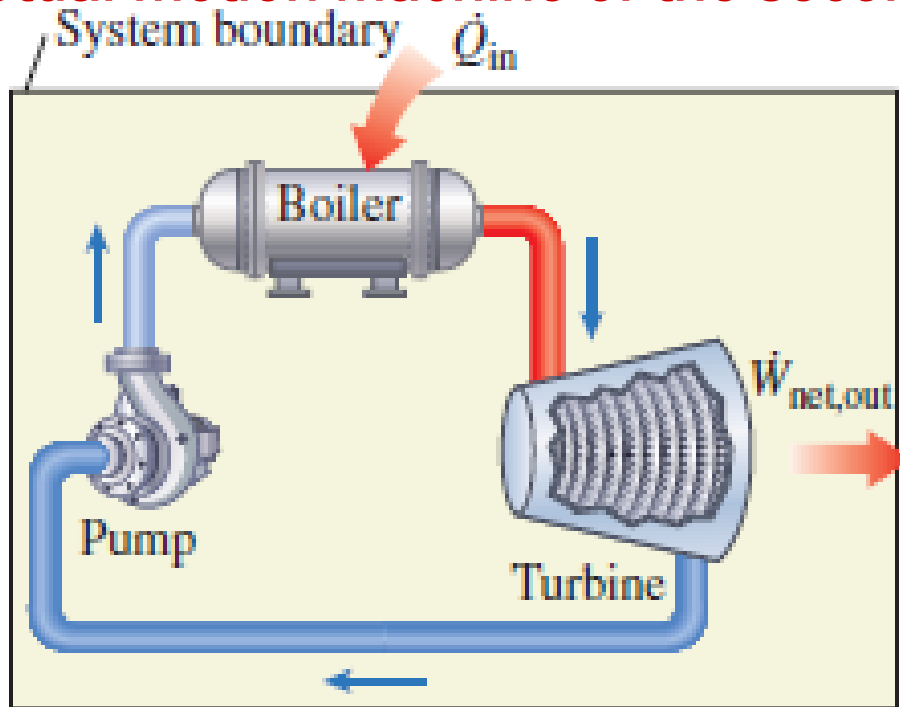
A device that violates the first law of thermodynamics (by *creating* energy) is called a **perpetual-motion machine of the first kind** (PMM1). It is proposed to heat the steam by resistance heaters placed inside the boiler, instead of by the energy supplied from fossil or nuclear fuels. Part of the electricity generated by the plant is to be used to power the resistors as well as the pump. The rest of the electric energy is to be supplied to the electric network as the net work output. The inventor claims that once the system is started, this power plant will produce electricity indefinitely without requiring any energy input from the outside.



A perpetual-motion machine that violates the first law of thermodynamics (PMM1).

Perpetual-Motion Machines of 2nd Kind

A device that violates the 2nd law of thermodynamics is called a **perpetual-motion machine of the second kind**

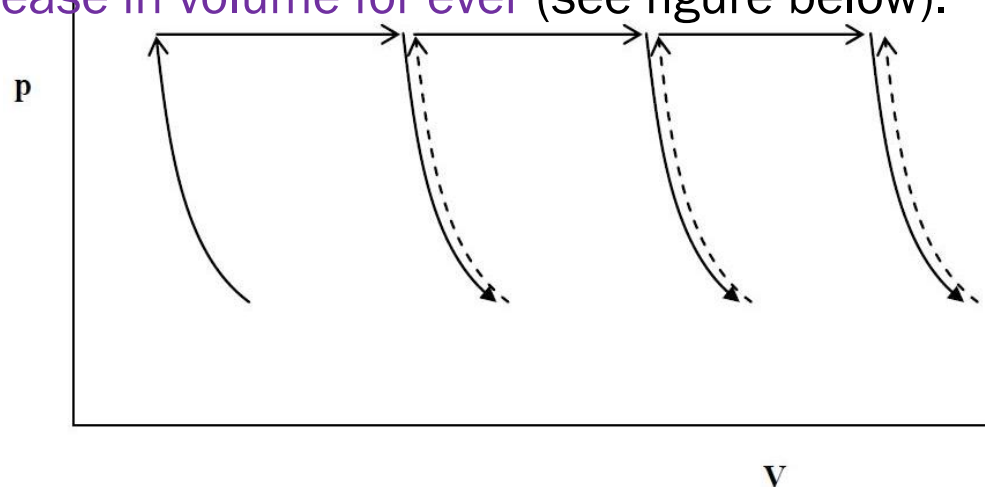


A perpetual-motion machine that violates the second law of thermodynamics (PMM2).

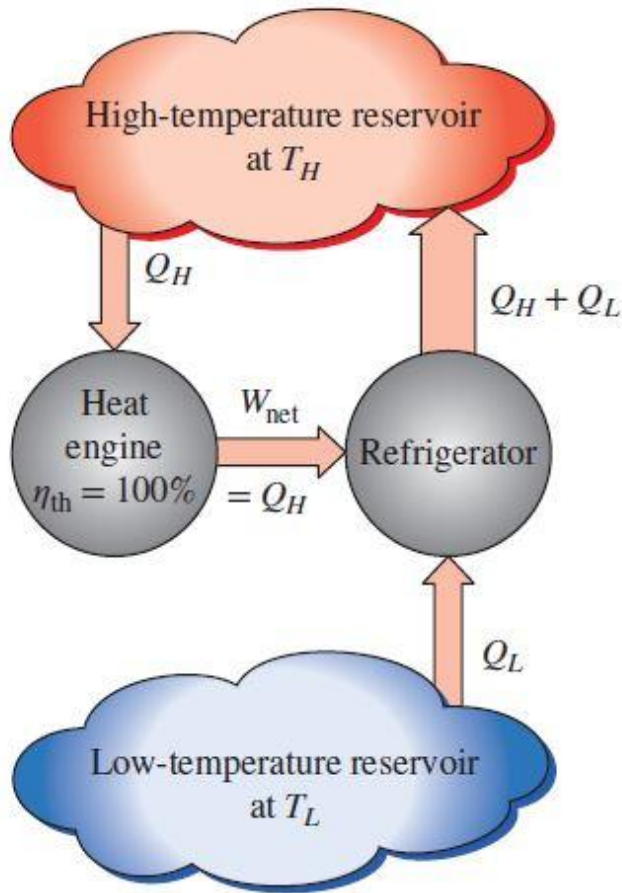
One inventor thought that more than one half the heat is lost in the condenser of the power plant. Hence, he suggested a modification by throwing this wasteful component, condenser and directly connecting the turbine exit to the pump inlet the efficiency of the engine can be improved theoretically to 100% and at least to 80% in actual power plant.

Do we really need a condenser where heat is rejected?

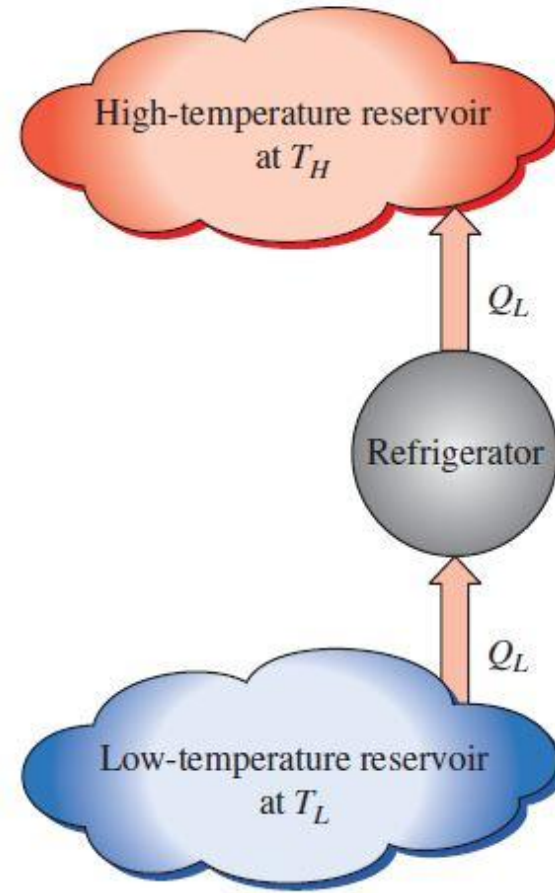
Let us look into the thermal power plant operation a little deeply. After expansion of the steam in the turbine for production of work, do we really need a condenser where heat is rejected? What will happen if we dispense with the condenser, compress the steam back (preferably by an adiabatic process) and send it back to the steam generator? There are two problems with this procedure (i) The work involved in compression is very less if it happens in liquid phase $\int v dp$, with the help of a pump because of the small liquid volume. If we try to compress vapor, the amount work involved will be enormous ($v_{vapour} \gg v_{liquid}$) and **almost the entire turbine work may get consumed in compression** (ii) **Compression alone cannot bring the working fluid back to its initial state for carrying out the cyclic process.** Without heat rejection in the condenser, the working fluid will undergo an unclosed process, with increase in volume for ever (see figure below).



Equivalence of Kelvin-Planck and Clausius Statement



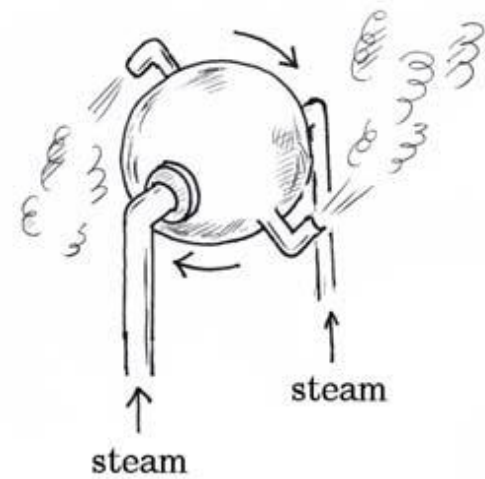
(a) A refrigerator that is powered by a 100 percent efficient heat engine



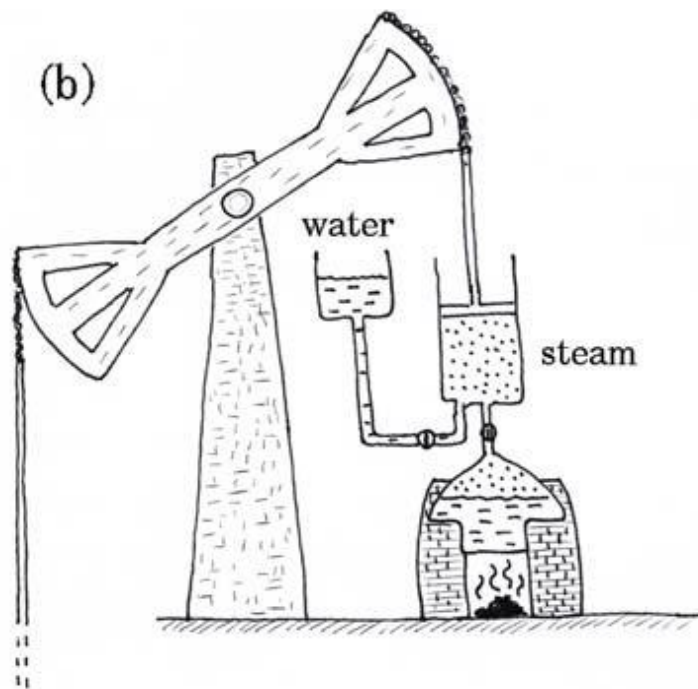
(b) The equivalent refrigerator

With $COP = \infty$

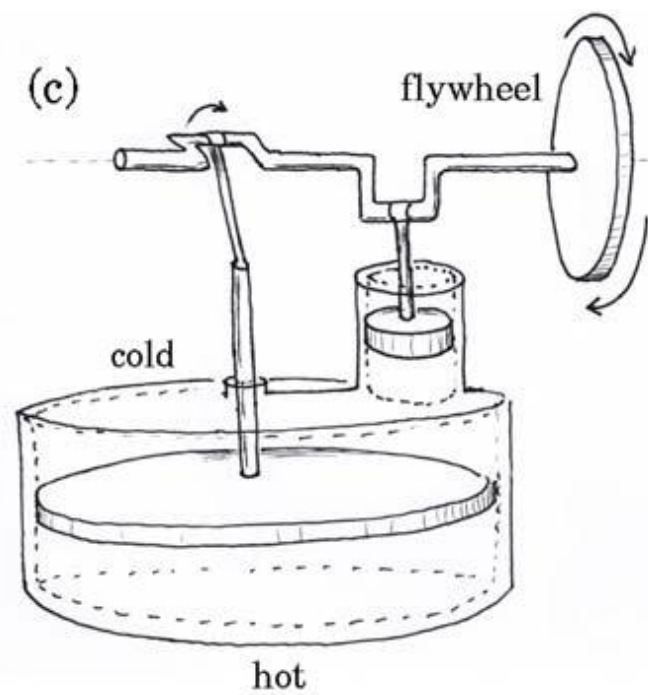
(a)



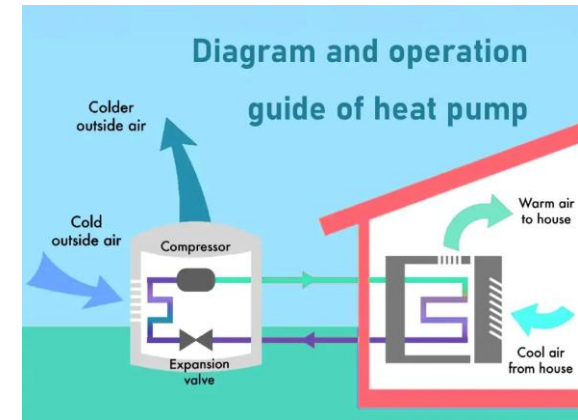
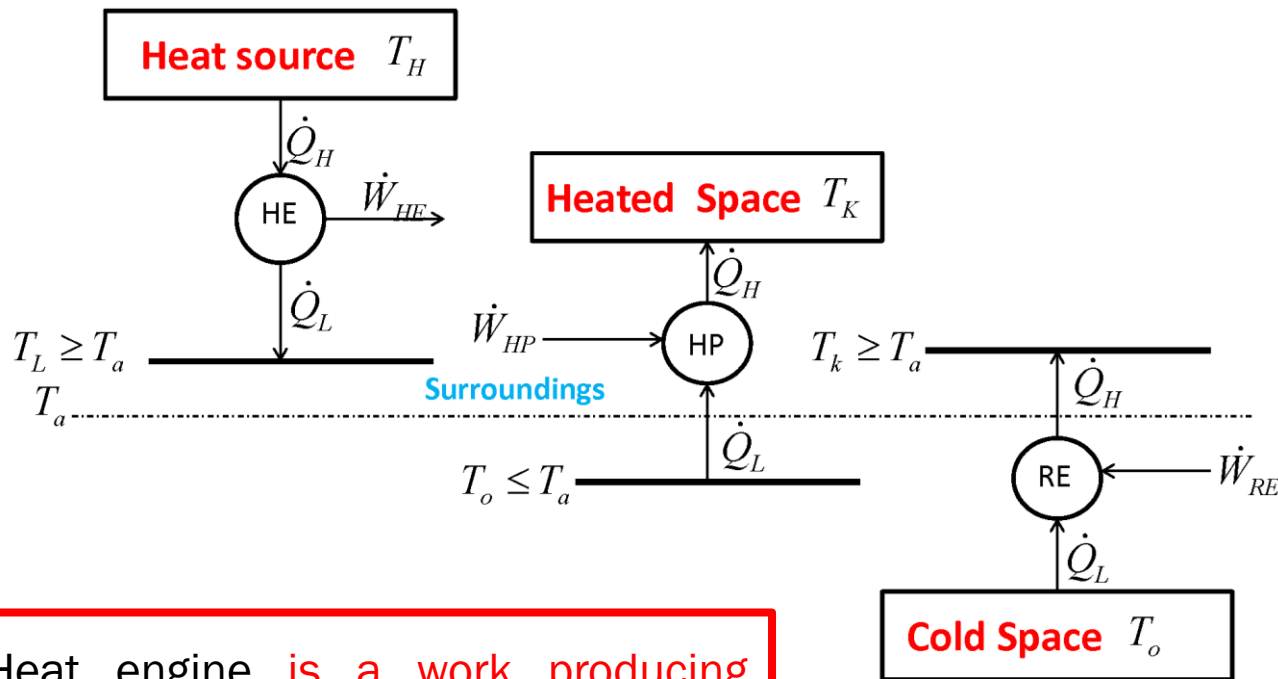
(b)



(c)



Heat engines, Refrigerators, Heat pumps:



Heat engine is a work producing machines operates in thermodynamic cycle for continuous production of power, under goes net positive heat and work interaction, may be closed system or a steady flow device. Interacts with at least two thermal reservoirs, the performance is defined in terms of efficiency.

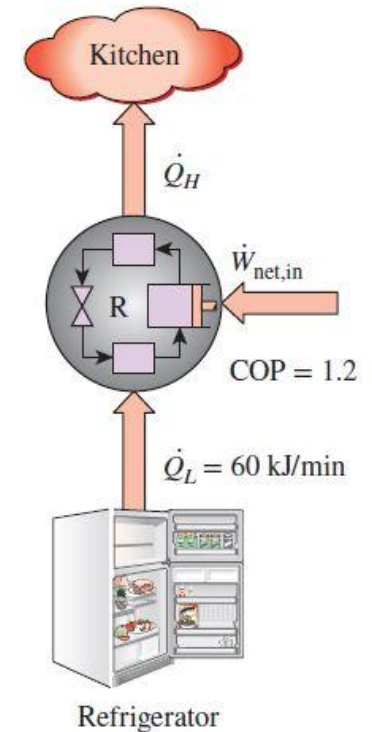
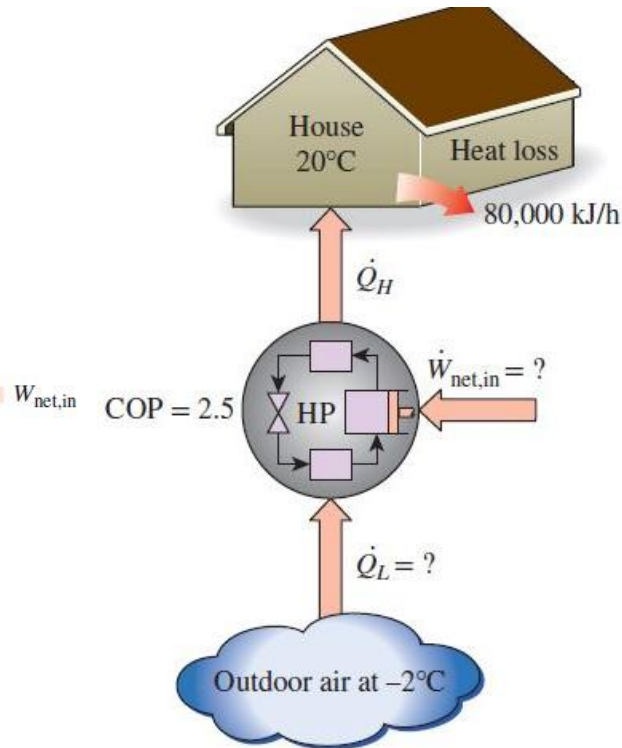
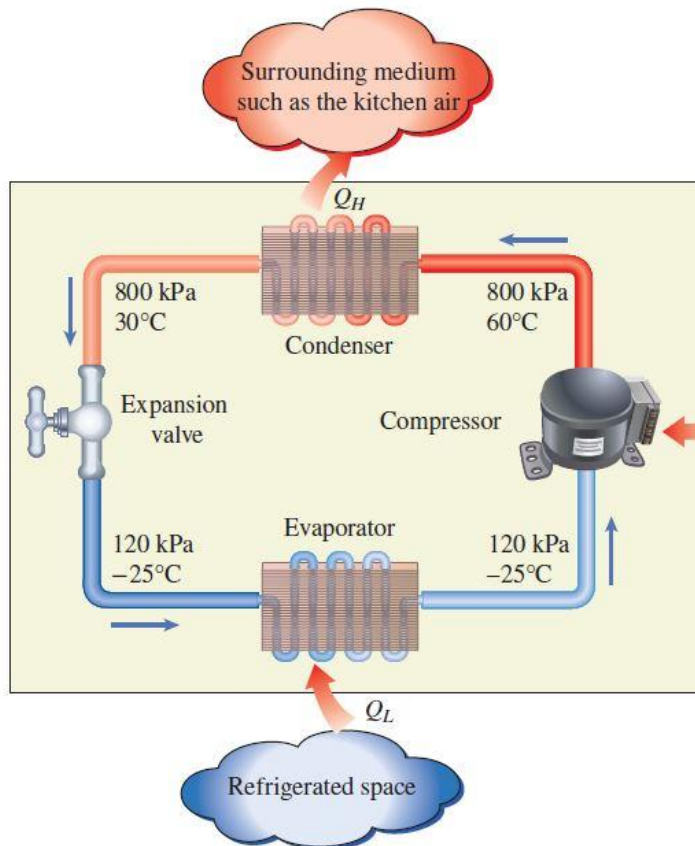
$$\eta_{HE} = \frac{W_{HE}}{Q_H} = \frac{Q_H - Q_L}{Q_H}$$

Thermodynamic devices such as Heat pump and refrigerators are the work consuming devices operates in thermodynamic cycle (reverse heat engine cycle) to produce continuous effect (Cooling/heating), undergoes net negative heat and work interaction. The performance is evaluated in terms of COP.

$$COP_{HP} = \frac{\dot{Q}_H}{\dot{W}_{HP}} = \frac{\dot{Q}_H}{\dot{Q}_H - \dot{Q}_L}$$

$$COP_{RE} = \frac{\dot{Q}_L}{\dot{W}_{RE}} = \frac{\dot{Q}_L}{\dot{Q}_H - \dot{Q}_L}$$

Thermodynamic Devices (Refrigerator and Heat Pump)

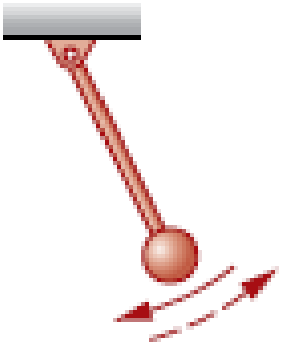


Reversible/irreversible Process

A process is **reversible** if, after it has occurred, **both the system and surroundings can by any means whatsoever be returned to their original states**. Any other process is irreversible

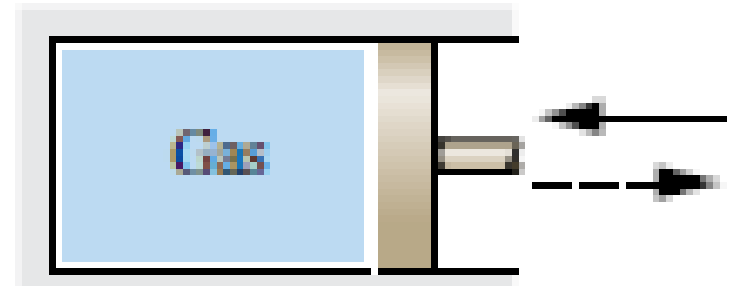
Reversible Process are important because:

- ✓ They provide the maximum work from work producing devices and minimum work input to the devices that absorb work to operate.
- ✓ For these devices and many other, reversible processes are standards of comparison
- ✓ A reversible process is an idealisation which can never be realised fully but which **is useful conceptually because it is easier to describe in mathematical terms than an irreversible process**.



To determine whether a process is reversible, it is necessary to apply 2nd law

A pendulum oscillating in an evacuated space. The pendulum motion approaches reversibility as friction at the pivot point is reduced.



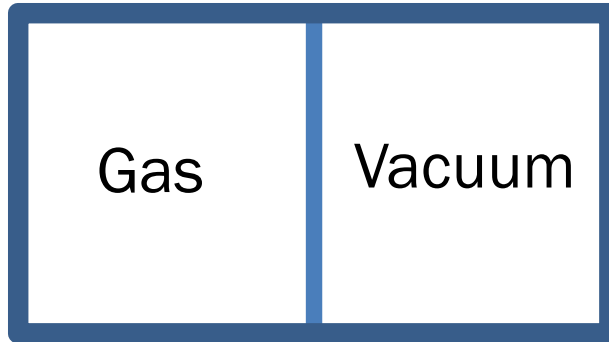
A system consisting of a gas compressed /expanded adiabatically in frictionless piston cylinder arrangement

Reversible/irreversible Process

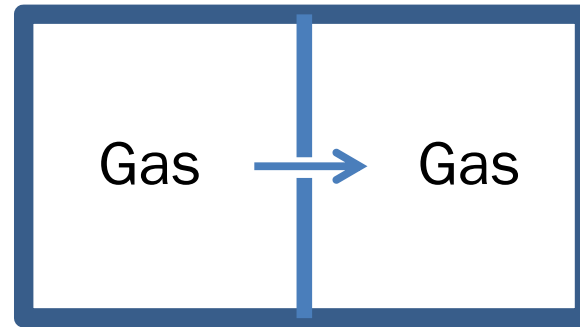
It is often easier to show that a process is irreversible than to show that one is reversible. Once certain irreversible phenomena are identified, we can often identify the reversible process by the absence of these irreversible phenomena. So how can you identify irreversible process? Recall that if a process is reversible, then both the system and surroundings can be restored to their initial state. However, if a process is irreversible, the reverse process is impossible.

Therefore, we can determine if a process is reversible by determining if the reverse process is possible. We can prove that a process is impossible as follows: (1) Assume that the process is possible. (2) Combine this process with other processes known from experience to be possible, to form a cycle violates the 2nd law. If such a cycle can be devised then the assumption of step one is false and the process in the question is impossible.

Demonstration that Free Expansion is Irreversible Process: An insulated tank is separated into two parts by a partition. A gas is held on one side of the partition; the other side of the tank is evacuated. This is the state A. An opening is then made in the partition and the gas expands to fill the entire tank. This is state B. No work is done as the gas expands.



state A



state B

$Q_{A-B} = 0$ (As the walls are insulated)

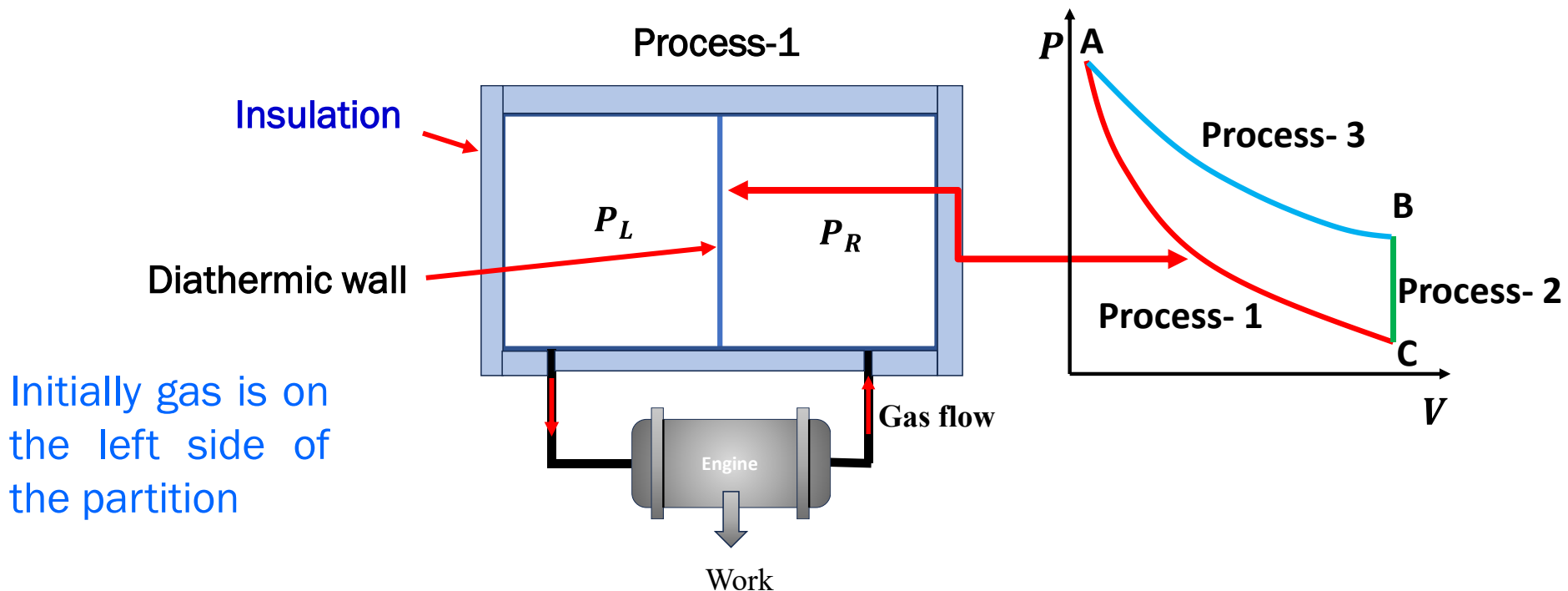
$W_{A-B} = 0$ (As the walls are rigid)

If you take gas as the system: $W_{A-B} = 0$ (No resistance)

Hence $U_A = U_B$ If ideal gas $T_A = T_B$

Consider the gas as the system, first assume that the reverse process is possible. The reverse process begins with the gas occupying the entire tank and requires the gas on one side of the partition to move against an increasing pressure to the other side of the tank. This must occur with no interaction with surroundings since the free expansion process occurred with no interaction with surrounding

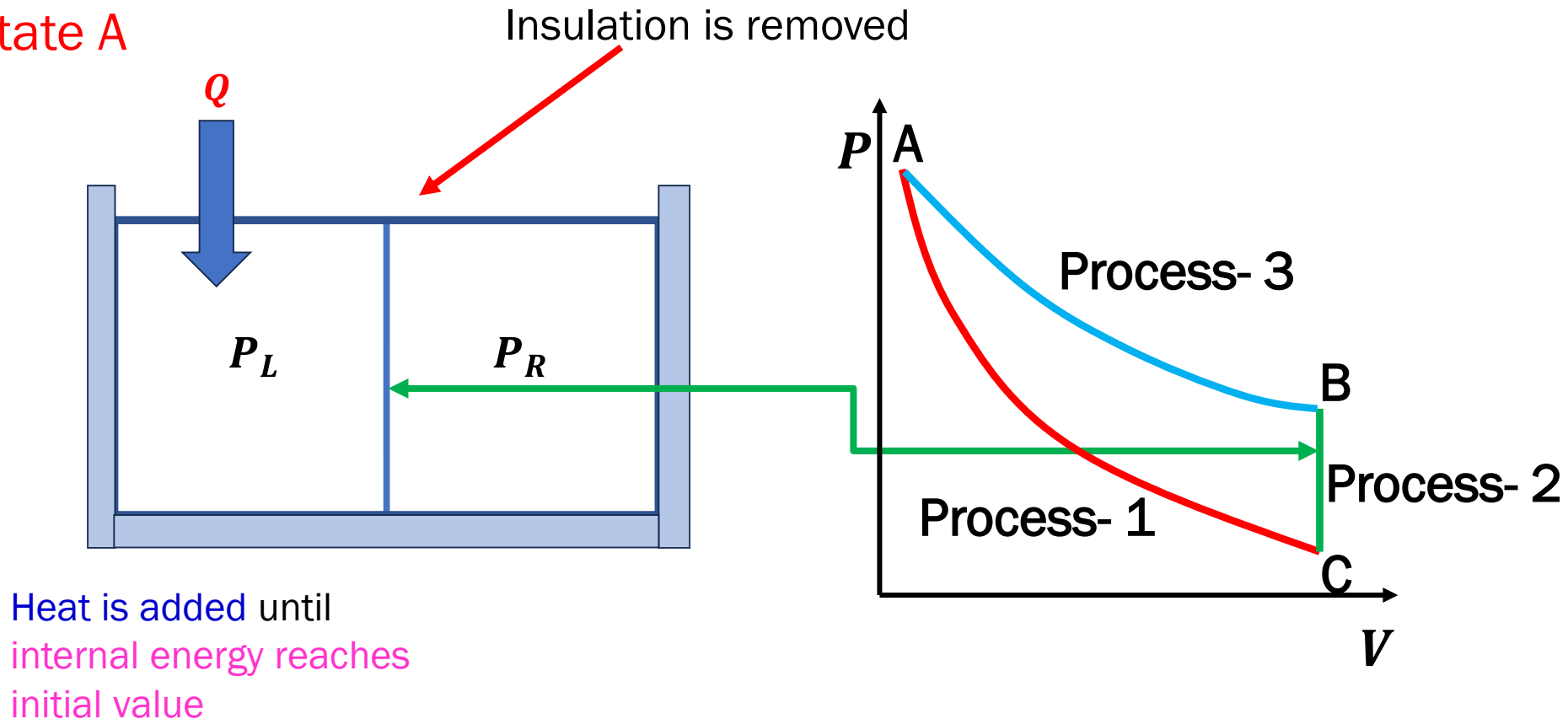
Process 1. Starting with gas all on one side of the partition, state A. Let part of it expand through an engine and into the other part of the tank until the pressure is same on the two sides of the partition. In expanding through the engine the gas does work, its internal energy drops. Let heat transfer occurs between the two parts of the tank so all of the gas comes to the same temperature, state C



Gas is going from state A to C following process 1 $W_{\text{Process-1}} = -\Delta U$

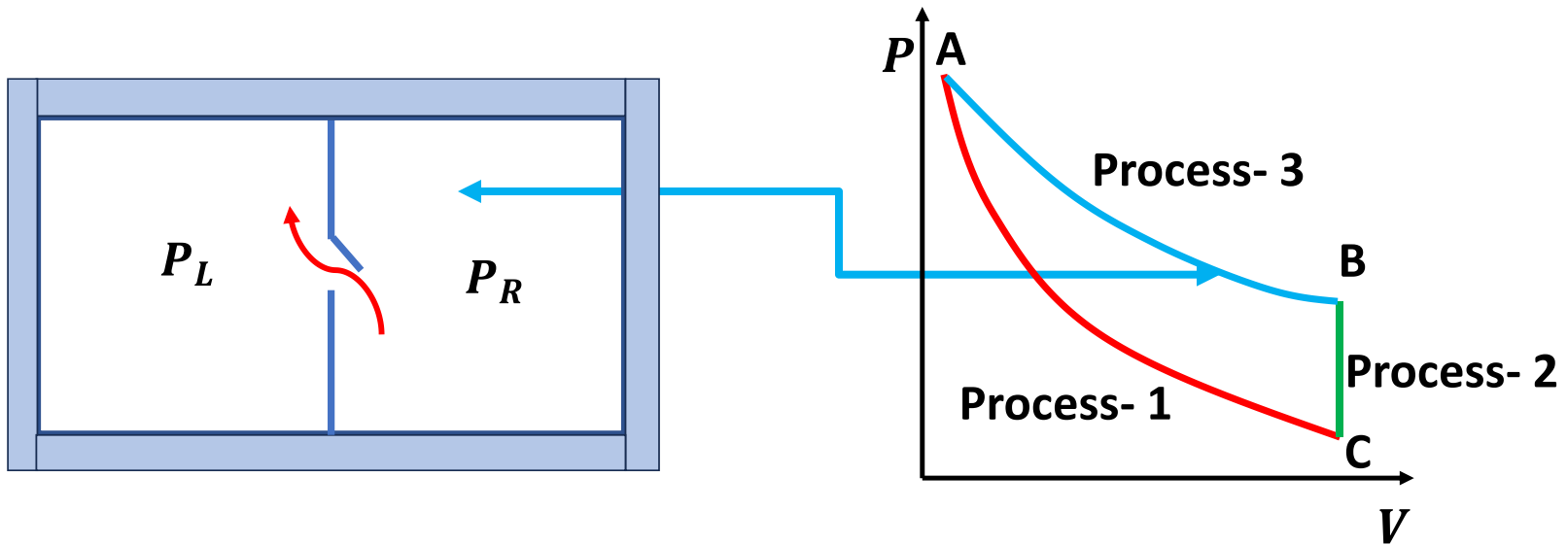
Gas Flows until $P_R = P_L$

Process-II. Remove the part of the tank insulation, and add heat from an external reservoir to the gas until its internal energy is restored to its initial value. The gas is now in state B with $U_B = U_A$. State B is also be the state of the gas would be in following a free expansion from state A



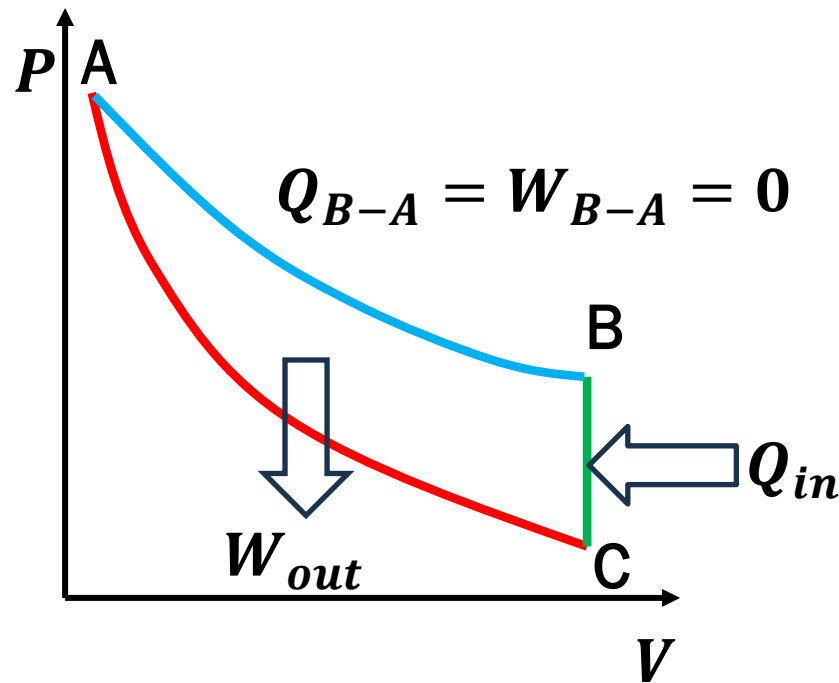
$$Q_{\text{Process-2}} = \Delta U, p_L = p_R$$

Process-3. Starting with the gas in state B, let the reverse of a free expansion occur to restore the system to its initial state A. Thus the cycle is completed. As in the beginning we have assumed the free expansion of the gas from state A-B is reversible, the reverse of that is the process from B-A must be possible, and follow the same path but in the reverse direction and finally all the gas must be restored to the left side that is state-A



Assumed to be possible) Gas flows as shown until $P_R = 0$, P_L increases to its initial value

This cycle results in a production of work while heat is absorbed from single reservoir. It hereby violates the second law of thermodynamics. Inspection of the cycle reveals that the process 1 and 2 are shown by experience to be possible, whereas process 3 is assumed to be possible. If the 2nd law is accepted then the assumption regarding process 3 is false. The reverse of a free expansion is therefor impossible and free expansion must be irreversible



Characteristics of Reversible Process

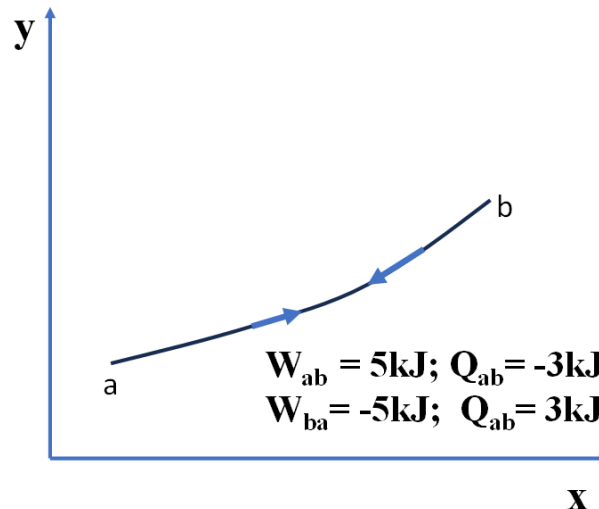
A process is said to be reversible if it satisfies the following characteristics

I. The process passes through infinite number of intermediate equilibrium points through infinitesimal gradient in properties

$$\Delta T; \Delta P; \Delta \mu$$

II. The process is quasistatic in nature and friction less.

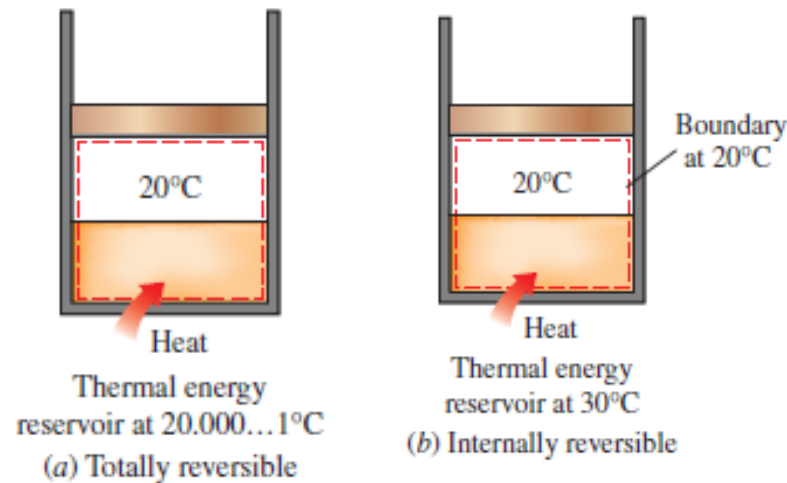
III. The reverse of a reversible process passes through the same intermediate equilibrium states. So during the reverse of a reversible process the magnitude of work and heat interaction will remain same as that in the forward process but of opposite sign (direction)



- IV. At the end of the reverse of the reversible process, at any intermediate equilibrium point, the system and surrounding are restored. (In a thermodynamic cycle the system is restored to its initial state but not the surrounding)
- V. At any intermediate point the process can be stopped.
- VI. The process which happens through finite gradient in properties are called irreversible processes.
- VII. An irreversible process is unidirectional does not possess any intermediate equilibrium state.
- VIII. The irreversible process is very fast and proceed with friction.
- IX. In the reverse of the irreversible process, the system and surroundings are not restored.
- X. All processes occurring in nature are spontaneous which is due to finite gradient in properties.

Internal & External Irreversibility

A process is irreversible if it involves heat transfer through finite temperature difference between the system and surrounding. However, the system can behave during this irreversible process just as though the heat were being transferred reversibly across an infinitesimal temperature difference. Such a process is called internally reversible because nothing occurs within the system to make it irreversible but it is externally irreversible.



A process meets the requirement of reversible process if it is both internally and externally reversible.

Friction less adiabatic and isothermal processes are both internally and externally reversible (V.I.M.P)